Distributed (Atomic) Commit

Material adapted from Lectures by Indy Gupta (IG) of UIUC

Reading

- Section 8.5 – Tanenbaum and Steen
- Section 17.3, 17.3.1 – Coulouris et al.

Aside: Distributed vs. Decentralized

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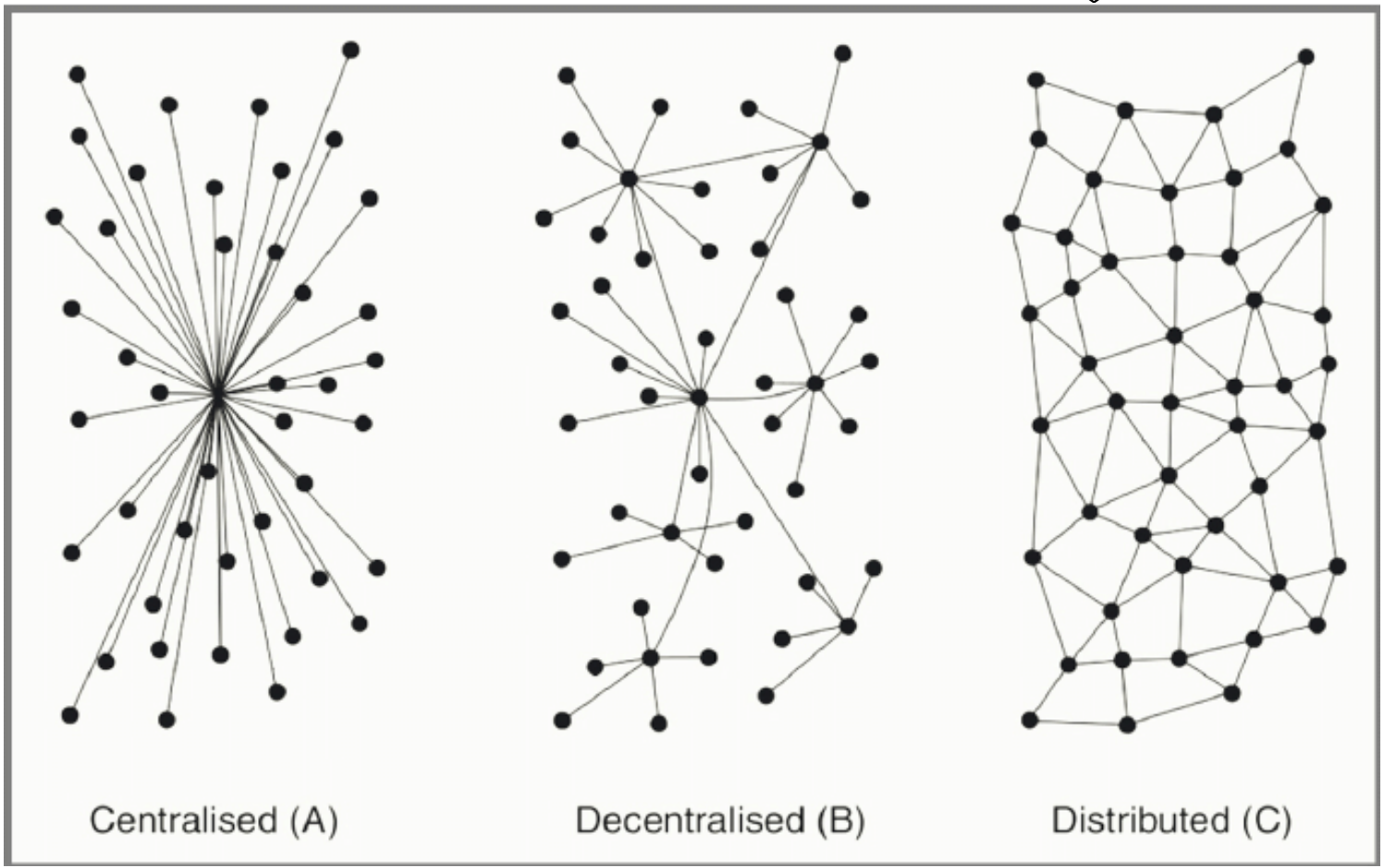

Centralized, decentralized and distributed network models by Paul Baran (1964)

What is Distributed Atomic Commit?

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- Encountered in many distributed systems and protocols
- Desired Outcome: either all entities commit to a transaction (e.g., database write) or all of them abort the transaction
  - Don't want some to execute the transaction while others abort
    - results in inconsistent system state
- Challenges: Nodes can fail/crash, messages can be lost
- Solved using protocols like 2-phase and 3-phase commit

1-Phase Commit

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- Coordinator sends a "commit/abort" message to all participants
- Participants "commit/abort" per coordinator message
- What if coordinator says "commit" but a participant is not able to commit?
  - Leads to inconsistent states.
- Need to allow participants the opportunity to inform they cannot commit

2-Phase Commit Protocol [Gray, 1978]

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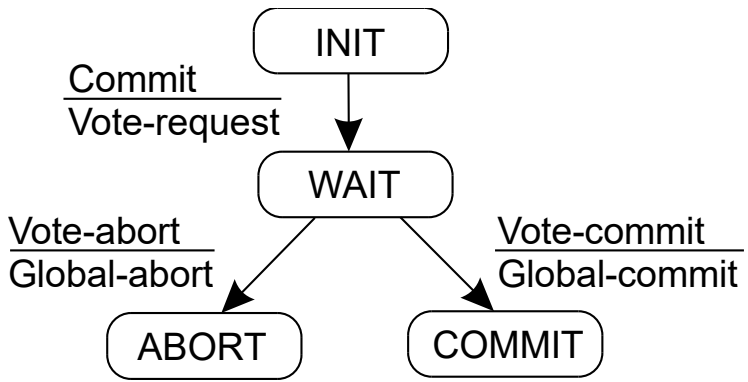
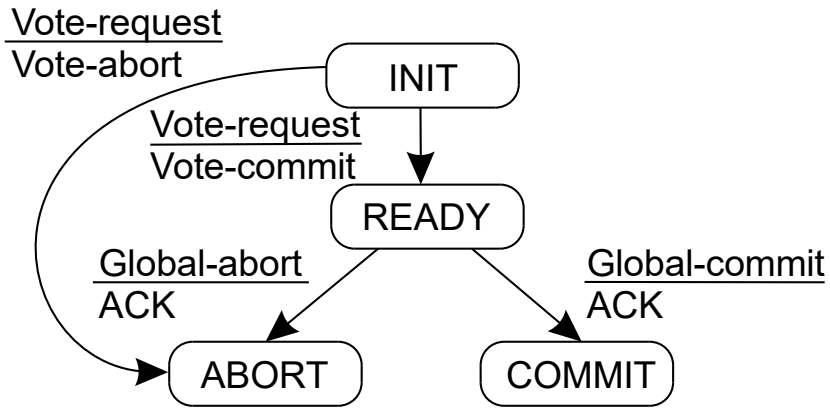
- Phase 1: Voting phase
  - Vote-request (coordinator) and Vote-commit/abort (participant)

- Phase 2: Commit phase
  - Global-commit/abort (coordinator) and commit/abort (participants)



Coordinator State Machine



Participant State Machine

2-Phase Commit Protocol – Failure Analysis

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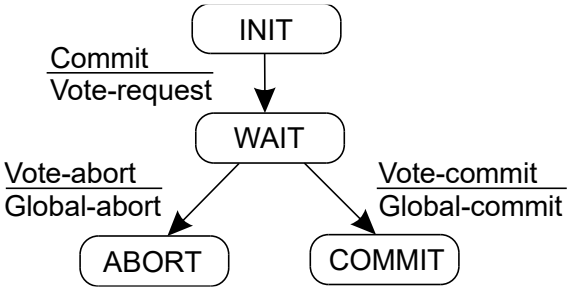

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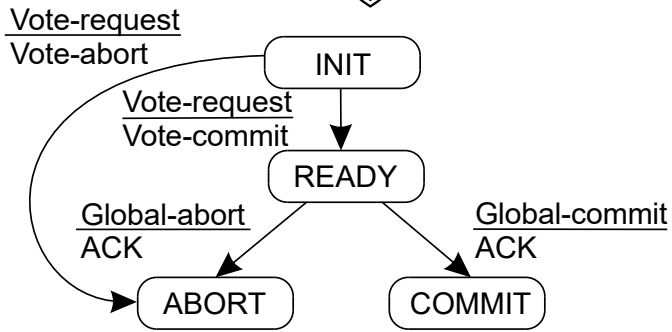

Participant State Machine

- Processes can be stuck in certain states when failures occur
  - e.g., Coordinator can be stuck in WAIT state when a participant crashes before voting

所有的參與者都會處於「懸掛」狀態。它們不敢提交（沒收到指令），也不敢回滾（大家都投了贊成），更不敢釋放資源鎖。整個系統的這部分資源一直被鎖死，直到協調者恢復。=> 同步阻塞

- Potential state where processes can be stuck
  - Coordinator: WAIT
  - Participants: INIT and READY

- Time-outs are used to make progress

2-Phase Commit Protocol – Failure Analysis

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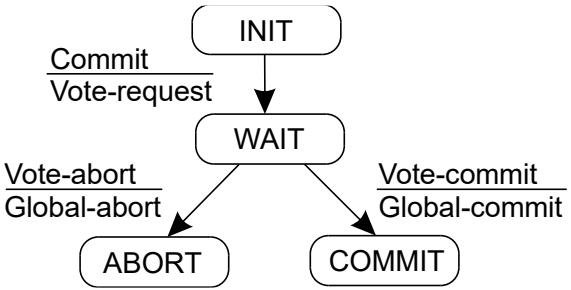

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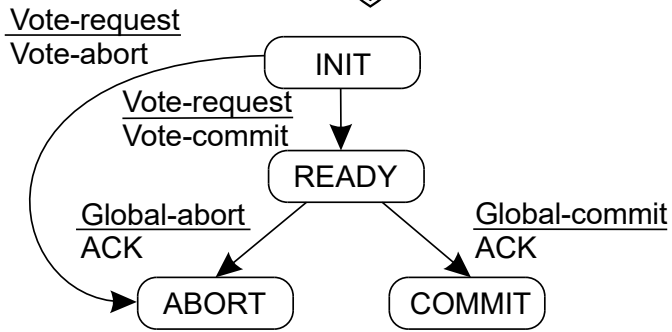

Participant State Machine

- Case 1: Coordinator "times-out" in WAIT state
  - Cause: Potentially a participant crashed or messages (vote-request and vote-abort/commit) were dropped
  - Action: Coordinator "aborts" and sends a Global-abort message

- Case 2: Participant "times-out" in INIT state
  - Cause: Potentially coordinator failed before sending vote-request or vote-request message got dropped
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2-Phase Commit Protocol – Failure Analysis

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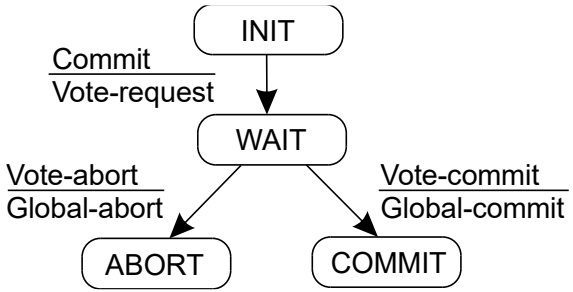
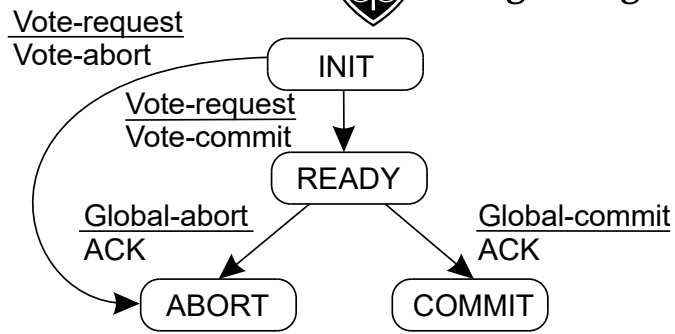


Coordinator State Machine



Participant State Machine

- Case 3: Participant P "times-out" in READY state
  - Cause: Potentially coordinator failed before sending a global commit/abort command or global commit/abort did not reach this participant (dropped)
  - Action: Participant P simply cannot abort as the rest might have committed the transaction.
    - Participant will have to ask another participant Q for their state and act accordingly
- NOTE: if P is in READY, possible states for other participants Are: INIT, ABORT, COMMIT, READY

State of Q	Action by P
COMMIT	Move to Commit
ABORT	Move to Abort
INIT	Move to Abort
READY	Contact another part.

3-Phase Commit Protocol

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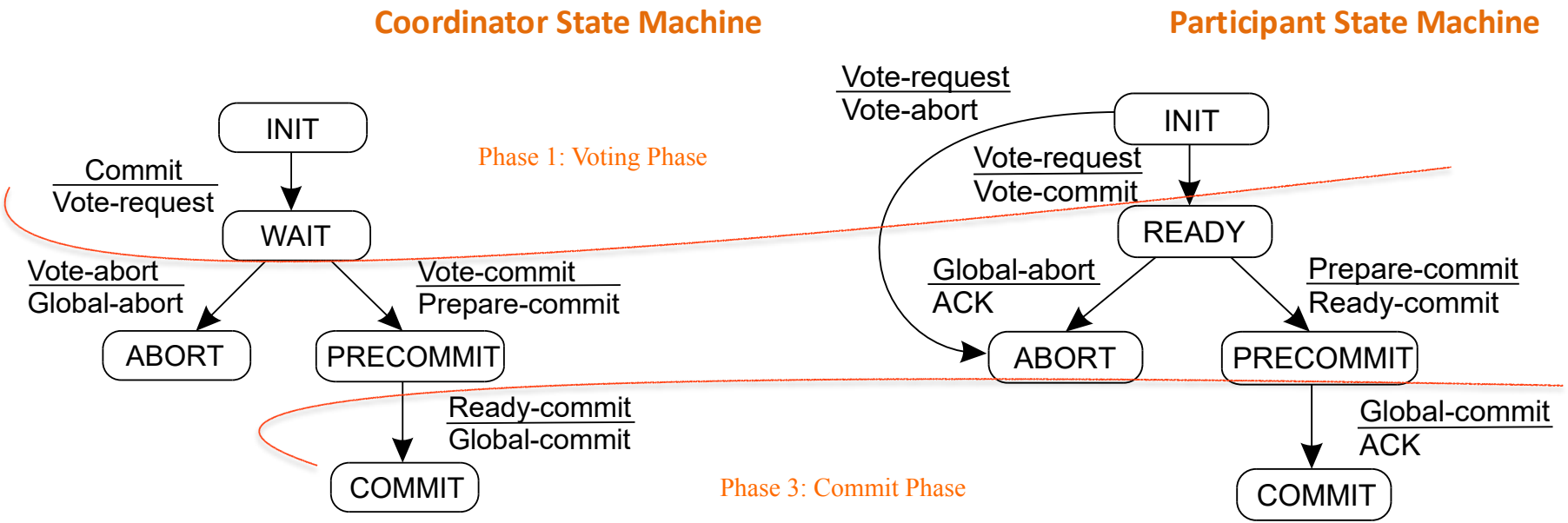

3-Phase Commit Protocol

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- Presented 3-PC satisfies the following:
  1. Synchronous within one state transition (i.e., any two participants cannot be more then one state apart)

     任意兩個參與者的狀態落差不會超過一個階段。

  2. It contains no state from which one can go to either a COMMIT or ABORT with one transition
     a. "PRECOMMIT" serves this purpose

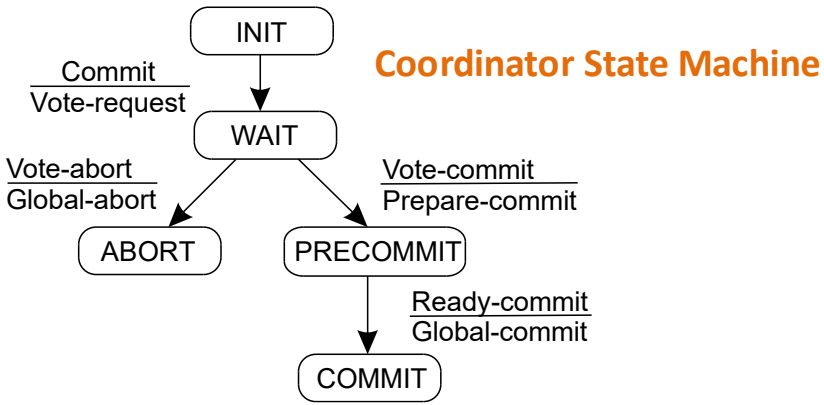
     有任何一個狀態可以「一步到位」直接跳到 COMMIT 或 ABORT

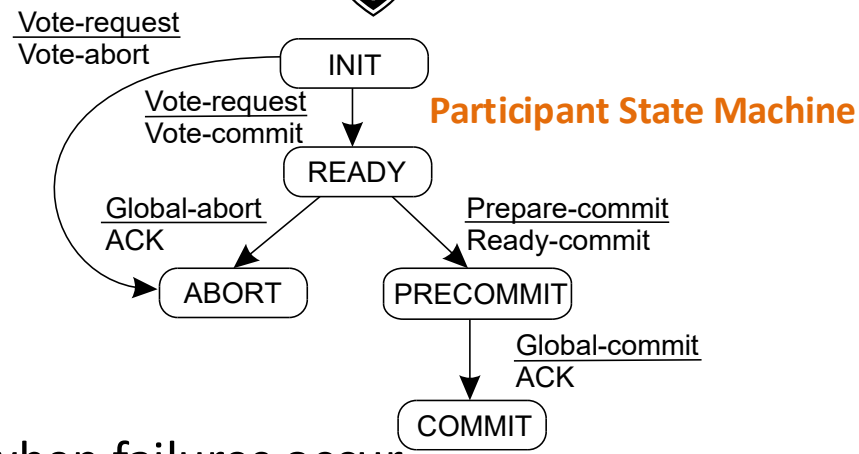
  3. It contains no "non-committable" (uncertainty about transaction still exists) state adjacent to a COMMIT state

     鄰近狀態（PRECOMMIT）已經排除了「不能提交」的可能性

- It can be shown that the above are necessary and sufficient to be a "non-blocking" protocol (certain failure model conditions apply)

3-Phase Commit Protocol



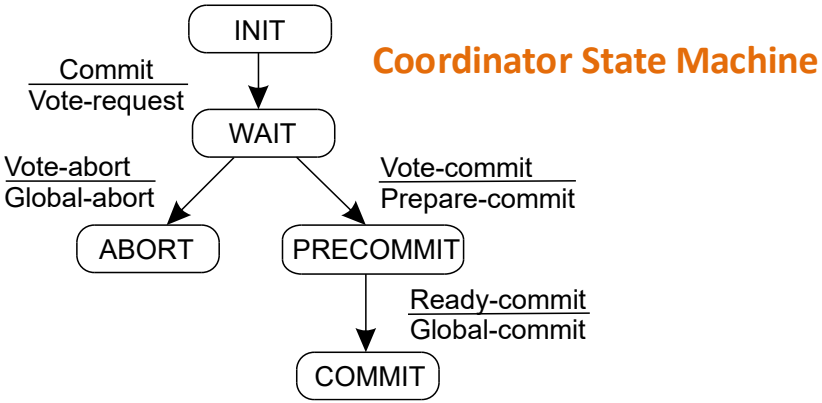


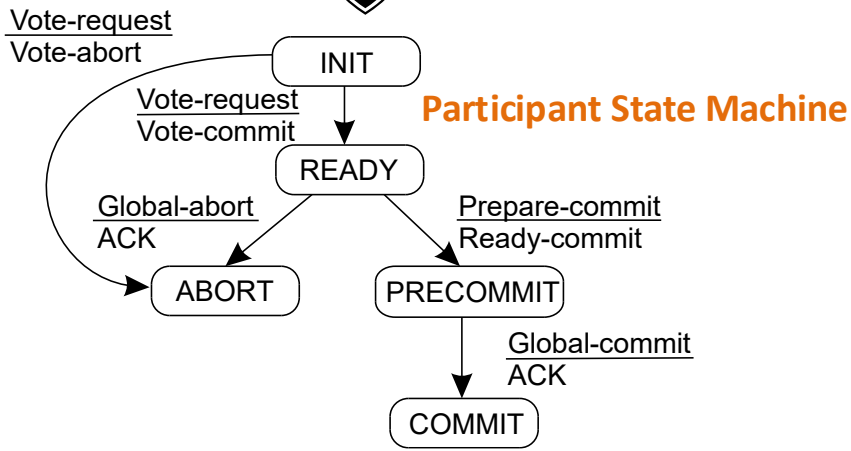
- Processes can be stuck in certain states when failures occur
  - e.g., coordinator can be stuck in WAIT state when a participant crashes before voting

- Potential state where processes can be stuck
  - Coordinator: WAIT and PRECOMMIT
  - Participants: INIT, READY, PRECOMMIT

- Time-outs are used to make progress

3-Phase Commit Protocol





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  - Action: Coordinator "aborts" and sends a Global-abort message

- Case 2: Participant "times-out" in INIT state
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3-Phase Commit Protocol

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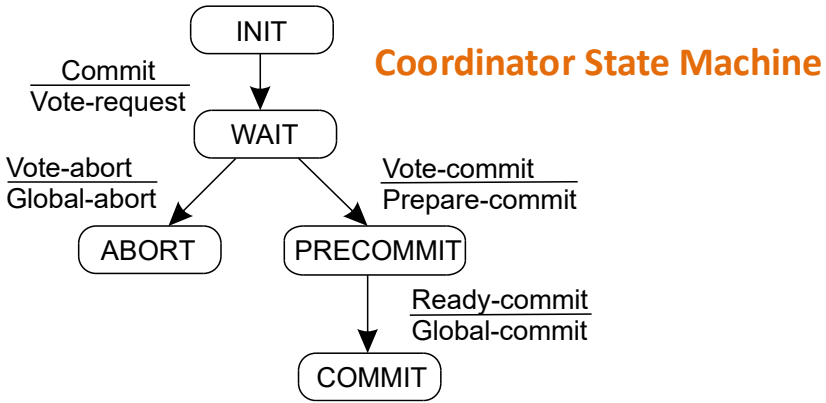

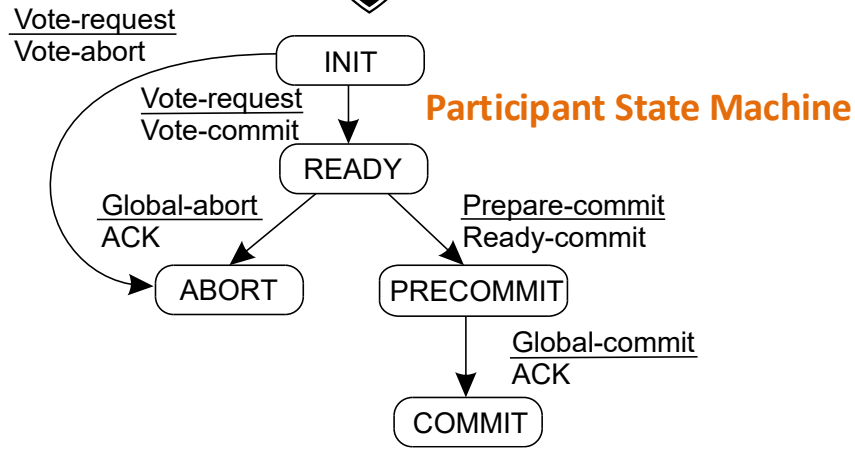

- Case 3: Coordinator "times-out" in PRECOMMIT state
  - Cause: Potentially a participant crashed after voting-to-commit or messages (Prepare-commit/ready-commit) were dropped
  - Action: Coordinator move to Commit and transmits "Global-Commit"
    - It counts on crashed process to recover and commit as it voted to commit

如果只有少數， commit，否則，abort

3-Phase Commit Protocol

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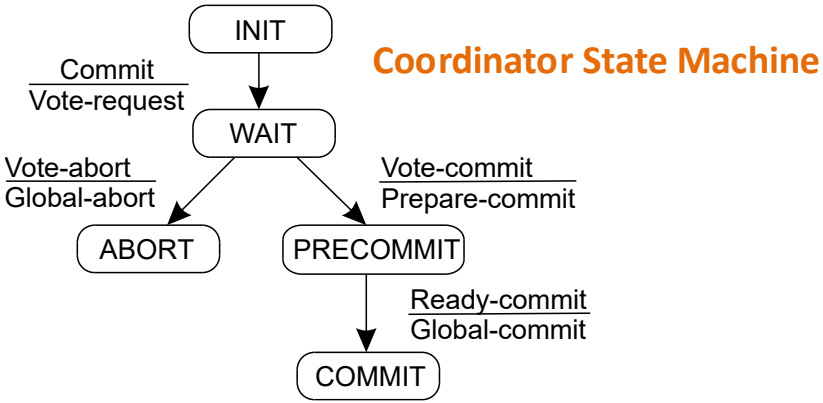

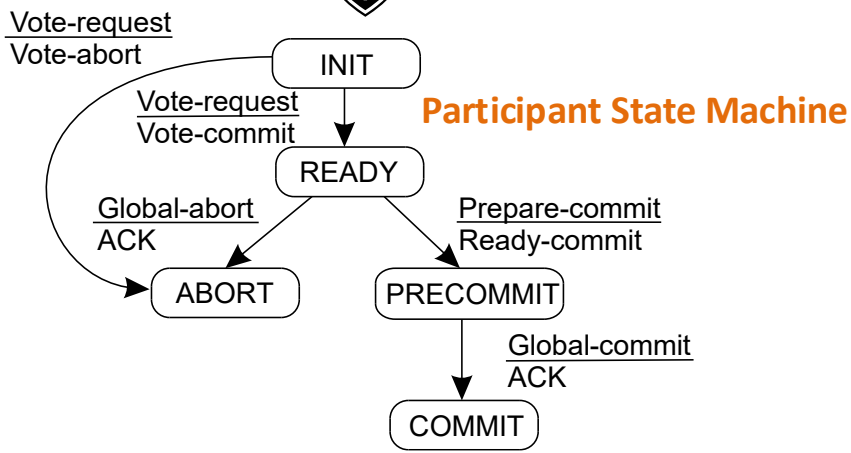

- Case 4: Participant "times-out" in READY state
  - Cause: Potentially coordinator failed before sending PREPARE COMMIT
  - Action: As in 2PC Participant contacts another participant Q
    - If Q is in ABORT P moves to that state as well
    - If Q is in INIT, both P and Q can move to ABORT
    - If all participants Q (contacted by P) are in READY state move to ABORT
    - If all participants Q (contacted by P) are in PRECOMMIT state move to COMMIT

- NOTE: If P is in READY state the only possible states for other participants are: INIT, READY, ABORT, PRECOMMIT

3-Phase Commit Protocol

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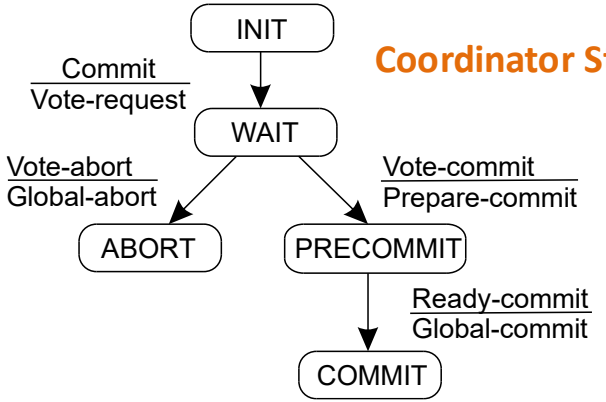
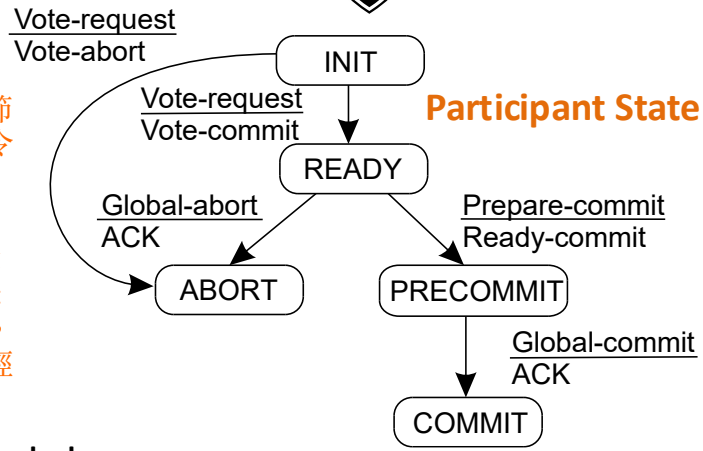


Coordinator State Machine

只要有任意一個節點收到了這個指令

並進入

PRECOMMIT 狀態，就足以證明：「在那個時間點，所有參與者都已經同意提交了。」



Participant State Machine

- Case 5: Participant "times-out" in PECOMMIT state
  - Cause: Potentially coordinator failed before sending GLOBAL COMMIT
  - Action: As in 2PC Participant contacts another participant Q
    - If Q is in COMMIT P moves to that state as well (and they form a majority)
    - If all participants Q (contacted by P) are in READY state move to ABORT
    - If all participants Q (contacted by P) are in PRECOMMIT state move to COMMIT

- NOTE: if P is in PRECOMMIT state the only possible states for other participants are: READY, PRECOMMIT, COMMIT

And coordinator will follow the result of participants decide

Questions?